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Absolutely. Here is the **enhanced version** of the Tier 1 & Tier 2 answers, keeping the **expanded 10-mark format, diagrams**, and now with **mnemonics embedded** for each topic so you can recall them instantly in the exam.

TIER 1 – BIG 6 (10-Mark Answers with Mnemonics)

1. Four stages of team development and growth (Tuckman)

 **Mnemonic:** For **S**ome **N**ice **P**erformance (Forming, Storming, Norming, Performing)

Bonus for 5th stage: After **P**erforming, **A**djourn (Adjourning)

Definition (1 mark):

A model describing how a team evolves from strangers to a high-performing unit through predictable stages.

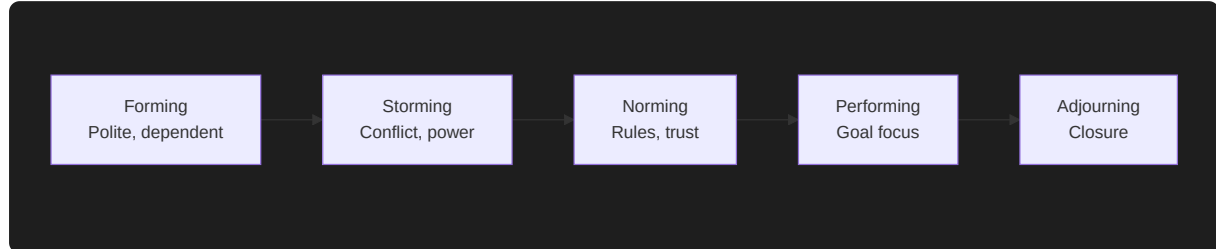
Detailed Answer (9 marks – 8-10 points):

Stage	Key Characteristics	PM Action Required
1. Forming	– Polite, anxious, dependent – Unclear goals – Individual roles undefined	– Provide clear direction – Introduce team members – Set initial ground rules
2. Storming	– Conflict over leadership – Clashing personalities – Subgroups form	– Facilitate conflict resolution – Reinforce common goals – Don't suppress – manage
3. Norming	– Rules and values emerge – Trust builds – Collaboration begins	– Step back gradually – Encourage autonomy – Celebrate small wins
4. Performing	– Focused on project goals – Flexible and adaptive – Minimal supervision needed	– Delegate authority – Remove external obstacles
5. Adjourning (optional)	– Team disbands – Closure and recognition needed	– Conduct lessons learned – Celebrate achievements

Text Diagram:

Forming → Storming → Norming → Performing → (Adjourning)
 (Anxiety) (Conflict) (Cohesion) (Results) (Closure)

Mermaid Diagram:



Exam Tip: Draw the performance dip during Storming, then rise through Norming to Performing.

2. Ways a project may be terminated + non-technical reasons

🧠 **Mnemonic for termination types:** Every Alien Is Scary (Extinction, Addition, Integration, Starvation)

🧠 **Mnemonic for non-tech reasons:** Some People Make Legal Rules (Strategy, Politics, Market, Legal, Regulatory)

Definition (1 mark):

Termination is formal end of project work. Non-technical reasons exclude technology failure.

4 Termination Ways (4 marks – 1 each):

Type	Description	Example	Resource handling
Extinction	Natural completion – success or failure	Product launched, team disbanded	Gradual release
Addition	Project becomes ongoing operation	IT project becomes helpdesk	Permanent reassignment
Integration	Resources absorbed into other projects	R&D team split into two new projects	Distributed
Starvation	Slow budget cutting – "death by thousand cuts"	Monthly budget reduced until zero	Demoralized attrition

Non-technical reasons (4 marks – 4 of these):

Reason	Example
Strategy change	Company pivots, project no longer aligns
Politics	Internal rivalry kills support
Market disappearance	Competitor launched similar product
Legal/Regulatory	New law makes project illegal
Sponsor loss	Champion leaves or loses power
Resource reallocation	Higher priority project emerges

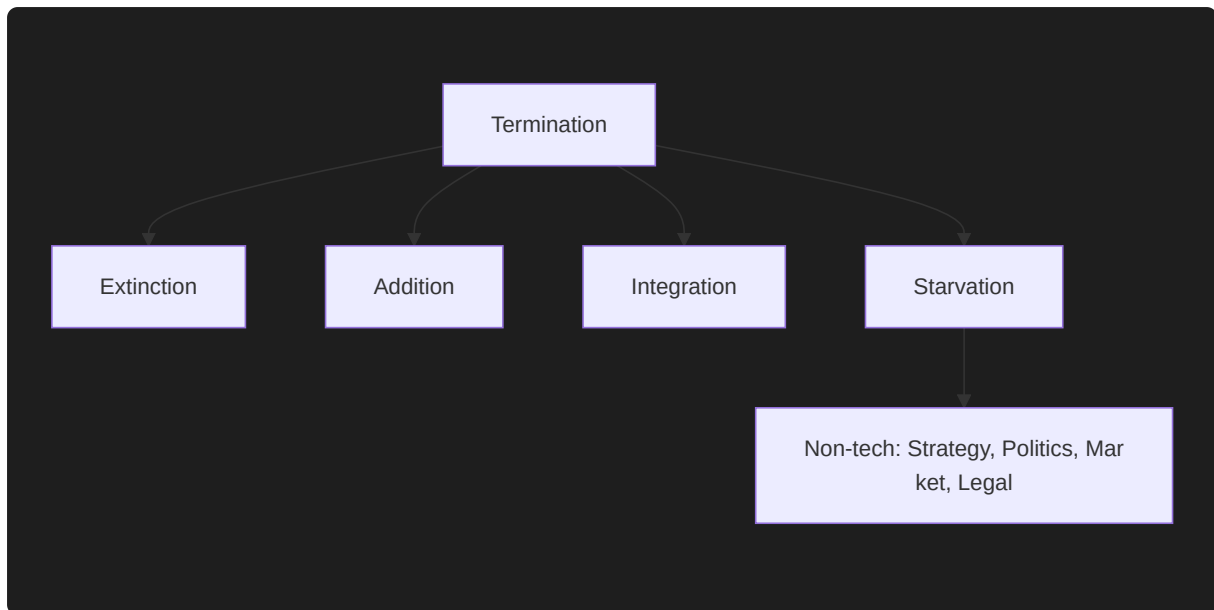
Text Diagram:

```

Project Termination
├─ Extinction (complete)
├─ Addition (becomes ops)
├─ Integration (absorbed)
└─ Starvation (slow death) ─ Non-tech reasons: Strategy, Politics, Market, Legal

```

Mermaid:



Exam Tip: Starvation is most common exam trap – it's NOT poor performance, it's political/funding.

3. Probability-impact matrix + risk response strategies

🧠 **Mnemonic for threats:** All Tigers Might Attack (Avoid, Transfer, Mitigate, Accept)

🧠 **Mnemonic for opportunities:** Every Shark Eats Anything (Exploit, Share, Enhance, Accept)

Definition (1 mark):

A qualitative risk analysis tool mapping probability vs impact to prioritize responses.

Part A: Probability-Impact Matrix (3 marks)

		IMPACT		
		Low	Med	High
Prob	High	Med	High	CRITICAL
	Med	Low	Med	High
	Low	Low	Low	Med

Part B: Threat Responses (3 marks)

Response	Action	Example
Avoid	Eliminate cause	Change scope to remove risky feature
Transfer	Shift impact	Insurance, fixed-price contract
Mitigate	Reduce probability/impact	Add testing, use proven tech
Accept	Acknowledge & budget	Contingency reserve

Part C: Opportunity Responses (3 marks)

Response	Action	Example
Exploit	Make it happen	Add resources to finish early
Share	Partner to capture	Joint venture
Enhance	Increase probability	Upskill team
Accept	Take if occurs	No proactive action

Mermaid:



Exam Tip: Draw the 3×3 grid first on rough sheet. Accept appears in both.

4. Differentiate between Functional, Pure Project, and Matrix organizations

Mnemonic: Functional = Functional boss; Project = Project boss; Matrix = Mixed boss (dual reporting)

Definition (1 mark):

Organizational structures defining project authority, resource control, and reporting lines.

Comparison Table (7 marks – 7 differences):

Factor	Functional	Pure Project	Matrix
PM authority	None / very low	High – full control	Low to high (weak/balanced/strong)
Resource control	Functional manager	Project manager	Shared (dual)
Team loyalty	Functional dept	Project only	Split (both)
Communication	Vertical silos	Horizontal	Both
Response time	Slow	Fast	Medium
Cost efficiency	High (no duplication)	Low	Medium
Best for	Routine projects	Large, unique projects	Cross-functional

Matrix subtypes (2 marks):

- **Weak Matrix:** Functional manager dominates
- **Balanced Matrix:** Shared authority
- **Strong Matrix:** PM dominates (near pure project)

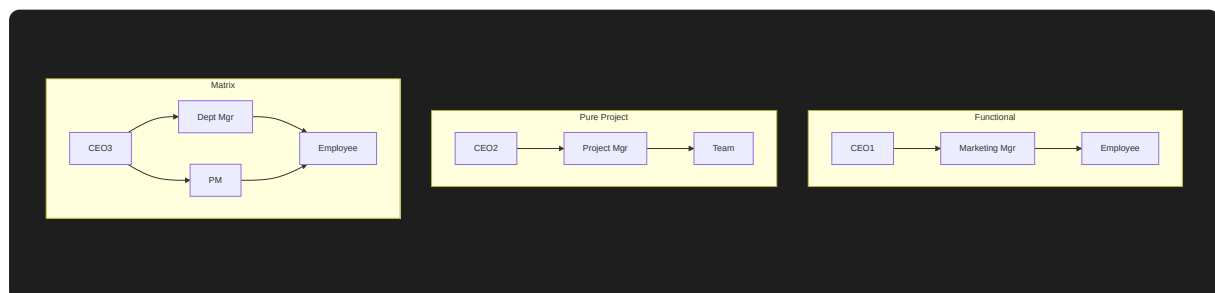
Text Diagram:

```

Functional:    CEO → Dept Mgr → Employee (no PM)
Pure Project:  CEO → PM → Team (full PM power)
Matrix:        CEO → Dept Mgr ↔ PM → Employee (dual reporting)

```

Mermaid:



Exam Tip: Always mention dual reporting as Matrix's defining pain point.

5. Responsibilities of project auditor + what makes a successful audit

🧠 **Mnemonic for responsibilities:** Real Very Cool Reports Required (Review, Verify, Check, Recommend, Report)

🧠 **Mnemonic for success factors:** I Can Ask Clearly (Independence, Clear scope, Access, Constructive)

Definition (1 mark):

Independent reviewer assessing project health, compliance, and processes – not individual performance.

5 Responsibilities (5 marks – R.V.C.R.R.):

Responsibility	Detailed Action
Review	Examine plans, schedules, budgets, risk registers
Verify	Compare actual vs planned (cost, schedule, scope)
Check	Assess compliance with standards, policies, regulations
Recommend	Propose corrective actions, process improvements
Report	Deliver findings to sponsor/steering committee (NOT to PM)

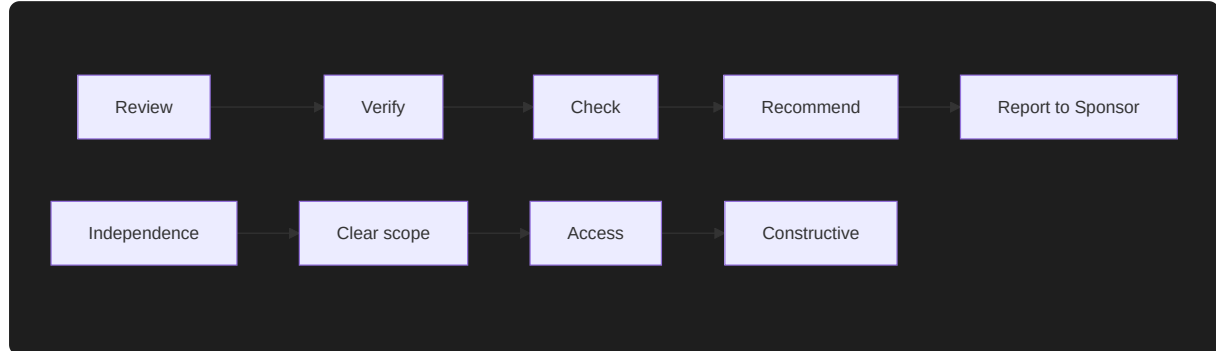
Success factors (4 marks – I.C.A.C.):

Factor	Why it matters
Independence	No reporting line to PM – avoids bias
Clear scope	Audit charter defines boundaries upfront
Access	Unrestricted documents, interviews, data
Constructive	Focus on fixing processes, not blaming people
Action taken	Recommendations implemented (bonus factor)

Text Diagram:

Audit Process: Input → Review → Verify → Check → Recommend → Report → Action
 (R) (V) (C) (R) (R)

Mermaid:



Exam Tip: Never say auditor reports to PM – that's a trap. Sponsor/steering committee only.

6. Goldratt's Critical Chain Method (CCM)

🧠 **Mnemonic:** Please Feed Rabbits – Project buffer, Feeding buffer, Resource buffer + No multitasking

Definition (1 mark):

A scheduling method considering both task dependencies AND resource constraints, adding buffers to protect project completion.

5 Key Components (5 marks):

Component	Explanation
Critical Chain	Longest path considering resource dependencies (not just logical like CPM)
Project Buffer	Time added at end – protects due date
Feeding Buffers	Where non-critical chains join critical chain – prevents delays spreading
Resource Buffers	Alerts that critical resource will be needed soon (warning, not time)
No multitasking	Focus on one task until done – eliminates context switching loss

Additional concepts (3 marks):

- **Student syndrome** – people start late; CCM removes safety time from tasks
- **Parkinson's Law** – work expands to fill time; CCM uses aggressive estimates + central buffers
- **Drum-Buffer-Rope** – Drum = critical chain pace; Buffer = protection; Rope = release timing

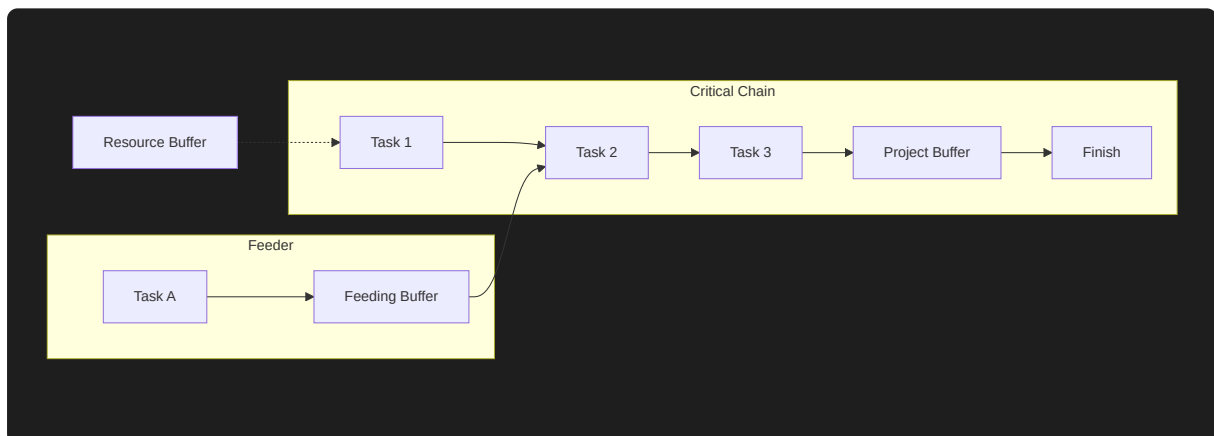
Text Diagram:

```

Non-Critical A ---[Feeding Buffer]---+
Non-Critical B ---[Feeding Buffer]---+
                                   v
Critical Chain → T1 → T2 → T3 → [Project Buffer] → Finish
               ↑
           [Resource Buffer: "Engineer needed"]

```

Mermaid:



Exam Tip: Key difference from CPM: CPM ignores resources; CCM includes them.

TIER 2 – STRONG 7 (10-Mark Answers with Mnemonics)

7. Work Breakdown Structure (WBS)

 **Mnemonic for WBS rules:** 100% Mutually Exclusive Deliverables Sized right (100% rule, Mutually exclusive, Deliverable-oriented, Sized 8-80 hours)

Definition (1 mark):

Deliverable-oriented hierarchical decomposition of total project scope into manageable work packages.

5 Key Rules (5 marks):

Rule	Explanation
100% Rule	Sum of children = parent level. No missing or extra work.
Mutually exclusive	No overlap between work packages.
Work package size	8–80 hours rule (or 1–2 weeks) – manageable.
Deliverable-oriented	Nouns, not verbs (e.g., "Foundation" not "Pour concrete")
Progressive elaboration	WBS evolves as scope clarifies.

WBS Levels Example (2 marks):

```

Level 1: House Construction
Level 2: Foundation, Structure, Roof, Finishing
Level 3: Foundation → Excavation, Rebar, Pour concrete
Level 4: Pour concrete → Order concrete, Setup forms, Pour, Cure
  
```

Benefits (2 marks):

- Basis for estimating cost, time, resources
- Assigns accountability
- Prevents scope creep
- Enables earned value management

Text Diagram:

```

House (L1)
├─ 1.0 Foundation (L2)
│   ├── 1.1 Excavation (L3)
│   └─ 1.2 Rebar (L3)
└─ 2.0 Structure (L2)
  
```

Exam Tip: Draw a 3-level WBS for any example (house, software, event).

8. Purchasing cycle

 **Mnemonic:** New Suppliers Never Offer Really Poor Coffee (Need, Select, Negotiate, Order, Receive, Pay, Close)

Definition (1 mark):

End-to-end process of acquiring goods/services from external suppliers.

7 Stages (7 marks – 1 each):

Stage	Activities	Document
1. Need	Requisition raised, specs defined	Purchase Requisition
2. Select	RFQ/RFP, evaluate bids	RFQ, Bid tabulation
3. Negotiate	Price, terms, delivery, quality	Negotiation notes
4. Order	Formal purchase order issued	Purchase Order (PO)
5. Receive	Goods received, checked vs PO	Receiving report
6. Pay	3-way match (PO, receipt, invoice)	Invoice, voucher
7. Close	Contract closure, feedback	Close-out report

3-Way Match (2 marks): PO + Receiving Report + Invoice must align before payment.

Text Diagram:

```
Need → Select → Negotiate → Order → Receive → Pay → Close
PR      RFQ      Terms      PO      Receipt  3-way  Archive
```

Exam Tip: Don't forget expediting – following up on late orders.

9. Communication planning and management

 **Mnemonic:** Who What How When Whom (5 Ws of communication planning)

Definition (1 mark):

Determining stakeholder information needs, methods, frequency, and responsibility.

5 Ws of Communication Plan (5 marks):

Element	Question
Who	Stakeholder / group
What	Type of information (status, risk, decision)
How	Channel (email, meeting, dashboard)
When	Frequency (daily, weekly, milestone)
Whom	Responsible sender

Key concepts (4 marks):

- **Communication channels:** $n(n-1)/2$ – as team grows, channels explode
- **Receiver responsibility:** Acknowledgment + confirmation of understanding
- **Noise:** Any barrier (language, culture, distance)
- **Feedback loop:** Confirmation message understood

Text Diagram:

```

Sender → Encode → Channel → Decode → Receiver
      ↑               ↓
      Noise ←———— Feedback

```

Exam Tip: Projects spend 90% of time communicating. Mention push vs pull communication.

10. Project charter – contents, who prepares & authorizes

🧠 **Mnemonic for contents:** Peter Often Makes Big Kites Selling At Markets – Purpose, Objectives, Milestones, Budget, Key deliverables, Scope, Assumptions, PM, Sponsor signature, Stakeholders (first letters)

Definition (1 mark):

Document that formally authorizes project and gives PM authority to apply resources.

Contents (6 marks – 6 of these):

Content	Details
Purpose	Business case, problem/opportunity
Objectives	SMART goals, KPIs, success criteria
High-level scope	Boundaries – what's in/out
Key deliverables	Major products/outputs
Milestone schedule	Key dates (not detailed)
Budget summary	Rough order of magnitude
Key stakeholders	Sponsor, PM, customer, team
Assigned PM	Name and authority level
Sponsor signature	Approves and funds
Assumptions/Constraints	Known limits

Who prepares & authorizes (2 marks):

- **Prepares:** Project Manager (or initiator/PMO)
- **Authorizes:** Project Sponsor (or steering committee)

Text Diagram:

```
Business Case → Charter (PM) → Review → Sign (Sponsor) → PM gets authority → Planning
```

Exam Tip: PM prepares but does NOT authorize. Sponsor authorizes.

11. Top-down vs bottom-up budgeting

 **Mnemonic:** Top-down = Total first; Bottom-up = Build from tasks

Definition (1 mark):

Budgeting approaches – top-down (executive driven) vs bottom-up (team driven).

Comparison Table (8 marks – 8 differences):

Factor	Top-down	Bottom-up
Start	Senior management sets total	Task-level estimates rolled up
Speed	Fast (hours)	Slow (days/weeks)
Accuracy	Rough, padded	High (if tasks known)
Buy-in	Low – imposed	High – ownership
Risk	May be unrealistic	May exceed funds
Best for	Early phases	Execution phase
Detail	Broad categories	Work package level
Bias	Political negotiation	Padding by team

Reconciliation (1 mark): Top-down sets target; bottom-up provides estimate; negotiate gap.

Text Diagram:

```

Top-down:      CEO sets $100k → Allocate to tasks
Bottom-up:     Task A $10k + Task B $20k = $45k → Present
Reconciled:    Target $50k vs Estimate $45k → Final $47k

```

Exam Tip: Never say one is "always better" – exam wants iterative reconciliation.

12. Contract + types of contracts

 **Mnemonic for contract types:** Fixed Cost Transfers Cost – FP (buyer safe), CR (seller safe), T&M (middle)

Definition (1 mark):

Legally binding agreement between buyer and seller specifying deliverables, price, terms.

Contract types with risk allocation (6 marks):

Contract	Buyer risk	Seller risk	When to use
Fixed Price (FP)	Low	High	Scope well defined
FP Incentive Fee	Low-Med	Medium	Cost savings shared
Cost Plus Fixed Fee	High	Low	R&D, uncertain scope
Cost Plus Incentive	Medium	Medium	Shared risk/reward
Time & Materials	Medium	Medium	Staff augmentation

Contract elements (2 marks): Offer, acceptance, consideration, scope, price, payment terms, termination clauses.

Text Diagram:

Contract Types

- ├ Fixed Price (buyer safe)
- ├ Cost Reimbursable (seller safe)
- └ Time & Materials (hybrid)

Exam Tip: FP = buyer happy (low risk); CR = seller happy (low risk).

13. Significance of IRR method

 **Mnemonic:** I Really Rank Projects (IRR = Internal Rate of Return, used to Rank Projects by percentage return)

Definition (1 mark):

Discount rate that makes NPV = 0. Measures project profitability as a percentage.

5 Points of Significance (5 marks):

Significance	Explanation
Profitability %	Easy to compare with cost of capital
Hurdle rate	IRR > required return → accept
Time value of money	Discounts future cash flows
Intuitive for mgmt	Managers think in percentages
Capital rationing	Rank projects by highest IRR

Limitations (3 marks):

- **Multiple IRRs** – Non-conventional cash flows give multiple solutions
- **Reinvestment assumption** – Assumes reinvestment at IRR (unrealistic)
- **Scale ignored** – Small project with 100% IRR may be worse than large project with 20% IRR

Comparison with NPV (1 mark): NPV theoretically superior (no reinvestment assumption). Use IRR as supplement.

Text Diagram:

Cash flows: -100, +60, +60

$NPV = -100 + 60/(1+r) + 60/(1+r)^2 = 0 \rightarrow IRR = 13.1\%$

Decision: If cost of capital = 10% → 13.1% > 10% → Accept

Exam Tip: Always mention – IRR ignores project size. Use NPV for absolute value.

Final Exam Cheat Sheet (Draw on Rough Sheet First)

Topic	Mnemonic	Keywords
Team stages	For Some Nice Performance	Forming, Storming, Norming, Performing
Termination	Every Alien Is Scary	Extinction, Addition, Integration, Starvation
Threats	All Tigers Might Attack	Avoid, Transfer, Mitigate, Accept
Opportunities	Every Shark Eats Anything	Exploit, Share, Enhance, Accept
Auditor	Real Very Cool Reports Required	Review, Verify, Check, Recommend, Report
Auditor success	I Can Ask Clearly	Independence, Clear scope, Access, Constructive
Goldratt	Please Feed Rabbits + No multitasking	Project, Feeding, Resource buffers
WBS	100% Mutually Exclusive Deliverables Sized	100% rule, Mutual exclusivity, Deliverable-oriented
Purchasing	New Suppliers Never Offer Really Poor Coffee	Need, Select, Negotiate, Order, Receive, Pay, Close
Comm plan	Who What How When Whom	Stakeholder, Info, Channel, Frequency, Sender
Charter	Peter Often Makes Big Kites Selling At Markets	Purpose, Objectives, Milestones, Budget, etc.
Budget	Top total, Bottom build	Top-down (fast, rough), Bottom-up (slow, accurate)
Contracts	Fixed Cost Transfers Cost	FP (buyer safe), CR (seller safe), T&M (hybrid)
IRR	I Really Rank Projects	Percentage return, compare to cost of capital

You now have:

- 10-mark expanded answers
- Text + Mermaid diagrams
- Mnemonics for every topic

Go write each twice. Good luck. 🚀